

ELP305: Expt No.1 - Interfacing with Arduino

Objective

Interfacing with Arduino

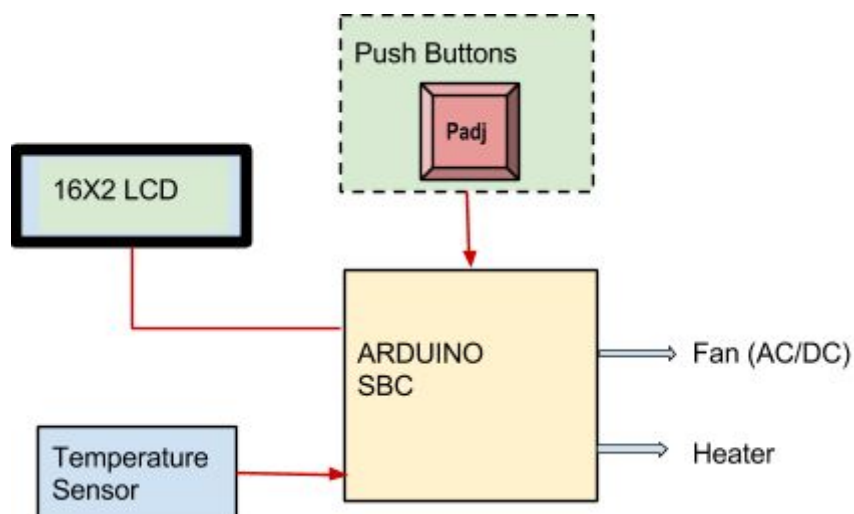
Problem Statement

Read ADC input from a temperature sensor and display output on LCD (Digital Thermometer) - then if the temp exceeds a certain point (a) turn on a LED (b) turn on a DC fan (c) turn on a AC fan. If the temp falls below a certain point, turn on a heater (resistor) . With on-off control, what is the level of precision (+/-) you can achieve if you use this system for temperature control of a heater ? How will you improve the precision (do the survey) ?

Stuff you will need to have

1. Arduino Board - www.arduino.cc
2. Temperature Sensor
3. 16X2 LCD - http://www.itron.com.cn/PDF_file/JHD162A%20SERIES.pdf
4. Push Buttons
5. Bread Board for prototyping

Schematic



Going further

1. Recording a daily temperature of a device. If it crosses the preset threshold then alarm.
2. Make a weather forecaster with an indicator of extreme temperatures.

References

- <https://www.arduino.cc/en/Tutorial/HelloWorld>

- <https://www.arduino.cc/en/Tutorial/LiquidCrystalDisplay>
- <http://www.ti.com/lit/ds/symlink/lm35.pdf>